illuminati - 2020

Advanced Physics Assignment-2

Liquids, Properties of Matter, Thermodynamics, SHM & Wave Motion

SECTION-1	SINGLE CORRECT ANSWER TYPE

1.(C) Vertical component of force of surface tension on liquid $= (2\pi r\cos\theta)\sigma$

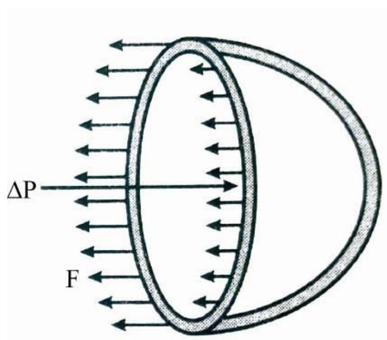
Work done by this force during capillary rise $= W = (2\pi\sigma r\cos\theta)\times h$

Where $h = \dfrac{2\sigma\cos\theta}{r\rho g}$ $\Rightarrow$ $W = (2\pi\sigma r\cos\theta)\times\dfrac{2\sigma\cos\theta}{r\rho g} = \dfrac{4\pi\sigma^2\cos^2\theta}{\rho g}$

Potential energy increase $= P = mg\left(\dfrac{h}{2}\right) = (\pi r^2 h)(\rho)\left(\dfrac{gh}{2}\right) = \dfrac{\pi\rho r^2 g h^2}{2}$

$\Rightarrow$ $P = \dfrac{2\pi\sigma^2\cos^2\theta}{\rho g}$ Heat liberated $= W - P = \dfrac{2\pi\sigma^2\cos^2\theta}{\rho g}$

2.(A) $\Delta V = V\theta\gamma, \dfrac{\Delta V}{V} = \gamma\theta$



Stress $= \left(\dfrac{\Delta V}{V}\right)B$

$\Delta P = \gamma\theta B$

$\dfrac{F}{\pi R^2} = \gamma\theta B$

$F = \pi\gamma\theta B R^2$

Tensile stress $= \dfrac{F}{2\pi R t} = \dfrac{\pi\gamma\theta B R^2}{2\pi R t}$

Tensile stress $= \left(\dfrac{\gamma\theta B R}{2t}\right)$

3.(A) Average kinetic energy with respect to space $= \dfrac{1}{3}kA^2$

Average potential energy with respect to space $= \dfrac{1}{6}kA^2$

Average kinetic energy with respect to time $= \dfrac{1}{4}kA^2$

Average potential energy with respect to time $= \dfrac{1}{4}kA^2$

4.(A) Rate of cooling of a body is the ratio of radiation power emitted by body and its heat capacity. As dimensions, surface material and temperature are equal the radiation power of both spheres will be same but heat capacity of hollow sphere is less so it will cool at a faster rate.

5.(C) $dp = -\rho g dh = -\dfrac{pm}{kT}\cdot g dh$

$\Rightarrow$ $\displaystyle\int_{p_0}^{p}\frac{dp}{p} = -\int_0^H \frac{mg}{kT}\cdot dh$ $\Rightarrow$ $p = p_0 e^{-\left(\frac{mg}{kT}\right)h}$

$$\begin{cases} m_\infty & = \text{mass till infinite length of column of area } A \\ m_h & = \text{mass till height } h \text{ of column column of area } A \end{cases}$$

$$P_0 = \frac{m_\infty g}{A}$$

$$P_h = \frac{(m_\infty - m_h)g}{A} = \frac{m_\infty g}{A} e^{-\frac{mgh}{kT}}$$

$$1 - \frac{m_h}{m_\infty} = e^{-\frac{mgh}{kT}}$$

$$\frac{m_h}{m_\infty} = \frac{n_h}{n_\infty} = 1 - e^{-\frac{mgh}{kT}}$$

6.(D) Pressure at S is less than atm pressure $P_S = P_0 - \rho g(H - h)$
So, mercury will not come out.

7.(C) As the volume of both are same, according to the thermal expansion concept, the change in volume is proportional to the total volume enclosed by the material hence it will be same for both spheres.

8.(D) Path difference $(\Delta x) = 50cm = \frac{1}{2}m$

$\Rightarrow$ Phase difference due to path difference $\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x \Rightarrow \phi = \frac{2\pi}{1} \times \frac{1}{2} = \pi$

Total phase difference $= \pi + \frac{\pi}{3} = \frac{4\pi}{3}$ $\Rightarrow$ $A = \sqrt{a^2 + a^2 + 2a^2 \cos(4\pi/3)} = a$

9.(B) On rising up as g decreases the time period of the pendulum increases and clock will slow down and start loosing time, time lost per unit time can be given as dT/T = (1/2)dg/g here g = GM//R^2 and verify that option **(B)** is correct.

10.(C) The maximum static frictional force is $f = \mu mg \cos\theta = 2\tan\theta\, mg \cos\theta = 2mg\sin\theta$

Applying Newton's second law to block at lower extreme position, we have $f - mg\sin\theta = m\omega^2 A$

$\Rightarrow$ $\omega^2 A = g\sin\theta$ As $\omega = \sqrt{\frac{k}{3m}}$ $\Rightarrow$ $A = \frac{3mg\sin\theta}{k}$

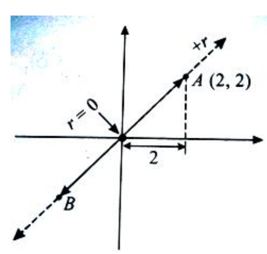
11.(C) At 4°C density of water is maximum so when heat is conducted and temperature of water at different regions become 4°C, it settles down at bottom and when temperature further falls below 4°C its density decreases and it will float above so 0°C will attain first at the region in between and at top.

12.(A) For chamber : $\frac{dQ}{dt} = k(\theta_1 - \theta_0) = k(\theta_2 - \theta_0)$ $\Rightarrow$ $\theta_1 = \theta_2$

For heater : $\frac{dQ}{dt} = e_1 A\sigma\left(T_1^4 - \theta_1^4\right) = e_2 A\sigma\left(T_2^4 - \theta_2^4\right)$ $\Rightarrow$ $\theta_1 = \theta_2$

$e_1 > e_2$ $\Rightarrow$ $T_1 < T_2$

13.(B) Let the line joining AB represent axis 'r'. By the conditions given 'r' coordinate of the particle at time t is



$r = 2\sqrt{2}\cos\omega t$ $\omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$

$\Rightarrow$ $r = 2\sqrt{2}\cos\pi t$ $x = r\cos 45° = \frac{r}{\sqrt{2}} = 2\cos\pi t$

$\Rightarrow$ $a_x = -\omega^2 x = -\pi^2 2\cos\pi t$ $\Rightarrow$ $F_x = ma_x = -4\pi^2\cos\pi t$

14.(D) As wavelength is more that means galaxy is receding away from earth at speed given as

$$v = \frac{\Delta\lambda}{\lambda} \times C = 0.005 \times 3 \times 10^8 = 1.5 \times 10^6 \ m/s$$

15.(D) Initially,

$$(V_{rms})^2 = \frac{\sum n_i v_i^2}{\sum n_i} = \frac{10\times(200)^2 + 20\times(400)^2 + ...}{100}$$

$$(V_{rms})^2 = (408\times1000)m^2s^{-2}$$

$$\Rightarrow \quad V_{rms} = 639\ ms^{-1}$$

So, temperature $T = \frac{1}{3}\left(\frac{mv_{rms}^2}{k_B}\right) = \frac{3\times10^{-26}\times4.08\times10^5}{3\times1.38\times10^{-23}} = 296\ K$

As, $v_{escape} = 900\ ms^{-1}$, so, all the particles with $v > v_{escape}$ or escape speed > $900\ ms^{-1}$ will be lost from the planet producing a cooling effect by evaporation.

After a long time when all molecules with $v > v_{escape}$ are lost, the rms speed of remaining molecules is

$$v_{rms}^2 = \frac{10\times(200)^2 + 20\times(400)^2 + 40\times(600)^2 + 20\times(800)^2}{90} = 342\times1000\ m^2s^{-2}$$

$$v_{rms} = 584\ ms^{-1}$$

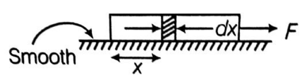
and so, temperature of planet $T = \frac{1}{3}\frac{mv_{rms}^2}{k} = 248K$

Hence, fall of temperature estimated $= 296 - 248 = 48K$

16.(A) $C = \frac{\Delta Q}{\Delta T} = \frac{\Delta U + \Delta W}{\Delta T}$

$\Delta U =$ same for both paths, $\Delta W_2 > \Delta W_1 \quad \Rightarrow \quad C_2 > C_1 \quad \Rightarrow \quad \frac{C_1}{C_2} < 1$

17.(D) Tension, $T = \frac{F}{L_0}.x$



Stress, $\sigma = \frac{T}{A} = \frac{F}{AL_0}x$

$$dU = \frac{1}{2}\cdot\frac{\sigma^2}{Y}Adx = \frac{1}{2}\cdot\frac{F^2}{A^2L_0^2}\cdot x^2\frac{A}{Y}dx \quad \text{or} \quad dU = \frac{F^2}{2AL_0^2Y}x^2dx$$

$$\Rightarrow \quad U = \frac{F^2}{2AYL_0^2}\int_0^{L_0} x^2dx \qquad U = \frac{F^2}{2AYL_0^2}\cdot\frac{L_0^3}{3} = \frac{F^2L_0}{6AY}$$

18.(D) Suppose that the temperature of the water in the first vessel is $\theta_1(t)$ and that of the second is $\theta_2(t)$, then

$$ms\frac{d\theta_1}{dt} = -\frac{KA}{L}(\theta_1 - \theta_2) \qquad ... (i)$$

and $$ms\frac{d\theta_2}{dt} = \frac{KA}{L}(\theta_1 - \theta_2) \qquad ... (ii)$$

from (i) and (ii), we get

$$\frac{d\theta}{dt} = \frac{-2KA}{msL}\theta, \text{ where } \theta = \theta_1 - \theta_2$$

The time, in which the temperature difference reduces to $1/e$ for its initial value, is given by

$$\Delta t = \frac{msL}{2KA} \quad \Rightarrow \quad K = \frac{msL}{2A\Delta t}$$

19.(B) When the piston moves downwards, the density of air trapped will increase. So, the buoyancy force due to air on the block will become slightly more and the cube will move up slightly. Similar argument holds good when the piston moves upwards.

20.(A) Work done by external agent = change in gravitational potential energy of sphere and water

$\Delta U_s = mgh$

$\Delta U_w = -mg(h-r)$ [As water from surface goes down to occupy the spherical region at the bottom]

$\therefore \quad w_{ext} = mgh - mg(h-r) = mgr$

SECTION-2	**LINK-COMPREHENSION TYPE**

21.(A) $\gamma_L = \gamma_S \Rightarrow \gamma_L = 3\alpha_S$

22.(A) $\dfrac{x}{l} = \dfrac{\rho_S}{\rho_L}$ . . . .**(i)**

$$\frac{x}{l(1+\alpha_S\Delta\theta)} = \frac{\rho_S(1+\gamma_L\Delta\theta)}{\rho_L(1+\gamma_S\Delta\theta)} \quad \text{. . . .(ii)}$$

From equation (i) and (ii) $\gamma_L - \gamma_S + \alpha_S = 0 \quad \Rightarrow \quad \gamma_L = 2\alpha$

23.(A) $\rho_S = \rho_L$

$$\Rightarrow \quad \frac{\rho_L}{1+\gamma_L\Delta T} = \rho_S \quad \Rightarrow \quad 1+\gamma_L\Delta T = \frac{\rho_L}{\rho_S}$$

$$\Rightarrow \quad 1+\gamma_L\Delta T = 2$$

$$T' - T = \frac{1}{\gamma_L} \quad \Rightarrow \quad T' = T + \frac{1}{\gamma_L}$$

24.(B) Since the vessel network is symmetrical, flow in a vessel of level *i* + 1 is half the flow in a vessel of level *i*.

In this way, we can sum the pressure differences in all levels : $\Delta P = \sum_{i=0}^{N-1} Q_i R_i = Q_0 \sum_{i=0}^{N-1} \dfrac{R_i}{2^i}$

Introducing the radii dependences yields $\Delta P = Q_0 \sum_{i=0}^{N-1} \dfrac{8\ell_i \eta}{2^i \pi r_i^4} = Q_0 \dfrac{8\ell_0\eta}{\pi r_0^4} \sum_{i=0}^{n-1} \dfrac{2^{4i/3}}{2^i 2^{i/3}} = Q_0 N \dfrac{8\ell_0\eta}{\pi r_0^4}$

Therefore $\quad Q_0 = \Delta P \dfrac{\pi r_0^4}{8N\ell_0\eta}$

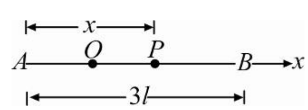
Hence, the flow rate for a vessel network in level *i* is $Q_i = \Delta P \dfrac{\pi r_0^4}{2^{i+3} N \ell_0 \eta}$

25.(C) Replace values in the formula and change units appropriately

$$Q_0 = \frac{\Delta P \pi r_0^4}{8N\ell_0\eta} = \frac{(55-30)\times 1.013\times 10^5 \times 3.1415 \times (6.0\times10^{-5})^4}{760 \times 48 \times 2.0\times 10^{-3} \times 3.5 \times 10^{-3}} = 4\times10^{-10}\ m^3/s$$

To obtain the final value in the requested units : $Q_0 \simeq 1.5\ m\ell/h$

26.(C) Take *x*-axis along *AB* and origin at *A*. Let $AP = x$, then $PB = (3l - x)$



The force on the particle of mass *m* at *P* is $\vec{F}$ given by

$$\vec{F} = 2\left(\frac{mg}{l}\right)(-\vec{x}) + \frac{mg}{l}\left(\frac{3l-x}{x}\right)\vec{x} = \left(3\,mg - \frac{3\,mg}{l}x\right) = \frac{3mg}{l}(l-x)$$

Hence, the equation of motion of mass m is: $m\dfrac{d^2x}{dt^2}=\dfrac{3\text{mg}}{l}(l-x)=m\omega^2(l-x)$

Where $\omega^2=3g/l$

This is a simple harmonic motion of period $T=2\pi\sqrt{\dfrac{l}{3g}}$.

The mean position is at the point $x=l$ (i.e.) at O, where $AO=l$.

Now, for point A, displacement $=-l$ and hence $v_A^2=\omega^2\left[a^2-(-l)^2\right]=\omega^2\left(a^2-l^2\right)$

Now, $v_A=3\sqrt{gl}$ (given) and hence $v_A^2=9gl=\dfrac{3g}{l}(a^2-l^2)$

$a^2=4l^2$ or $a=2l$

27.(B) $OB=2l$. O is the mean-position, B is the amplitude point. Also $|AP|=l=\dfrac{a}{2}$

Hence, the time t from A to O is given by $\dfrac{a}{2}=a\sin\omega t$

Given $\omega t=\dfrac{\pi}{6}\quad\Rightarrow\quad t=\dfrac{\pi}{6\omega}=\dfrac{\pi}{6}\times\dfrac{T}{2\pi}=\dfrac{T}{12}$

Hence, time from A to $B=\dfrac{T}{12}+\dfrac{T}{4}=\dfrac{T}{3}$

28.(A) Since, B is the amplitude point, the instantaneous velocity at $B=vB=0$.

29.(A) Let $p(z)$ denote the pressure of air at height z ; then, according tone of the assumptions $p(z)=p(0)$ $\rho_{Air}gz$, where $p(0)$ is the atmospheric pressure at zero altitude.

Throughout the chimney the Bernoulli law applies, that is, we can write

$$\frac{1}{2}\rho_{Smoke}v(z)^2+\rho_{Smoke}gz+p_{Smoke}(z)=\text{const.},\qquad (1)$$

where $p_{Smoke}(z)$ is the pressure of smoke at height z, ρ_{Smoke} is its density, and $v(z)$ denotes the velocity of smoke ; here we have used the assumption that the density of smoke does not very throughout the chimney. Now we apply this equation at two points, (i) in the furnace, that is at point $z=-\varepsilon$, where ε is a negligibly small positive number, and (ii) at the top of the chimney where $z=h$ to obtain

$$\frac{1}{2}\rho_{Smoke}v(h)^2+\rho_{Smoke}gh+p_{Smoke}(h)\approx p_{Smoke}(-\varepsilon)\qquad (2)$$

On the right hand side we have used the assumption that the velocity of gases in the furnace is negligible (and $-\rho_{Smoke}g\varepsilon\approx 0$).

We are interested in the minimal height at which the chimney will operate. The pressure of smoke at the top of the chimney has to be equal or larger than the pressure of air at altitude h ; for minimal height of the chimney we have $p_{Smoke}(h)\approx p(h)$. In the furnace we can use $p_{Smoke}(-\varepsilon)\approx p(0)$. The Bernoulli law applied in the furnace and at the top of the chimney (Eq. (2)] now reads.

$$\frac{1}{2}\rho_{Smoke}v(h)^2+\rho_{Smoke}gh+p(h)\approx p(0)\qquad (3)$$

From this we get : $v(h)=\sqrt{2gh\left(\dfrac{\rho_{Air}}{\rho_{Smoke}}-1\right)}$ (4)

$$v(h)\geq\frac{B}{A}\qquad (5)$$

Form which we have $h \ge \dfrac{B^2}{A^2}\dfrac{1}{2g}\dfrac{1}{\dfrac{\rho_{Air}}{\rho_{Smoke}}-1}$ (6)

30.(C) We can treat the smoke in the furnace as an ideal gas (which is at atmospheric pressure $P(0)$ and temperature T_{smoke}. If the air was at the same temperature and pressure it would have the same density according to our assumptions. We can use this to relate the ratio ρ_{Air}/ρ_{Smoke} *to* T_{Smoke}/T_{Air}, that is,

$$\frac{\rho_{Air}}{\rho_{Smoke}}=\frac{T_{Smoke}}{T_{Air}}, \text{ and finally} \quad (7)$$

$$h \ge \frac{B^2}{A^2}\frac{1}{2g}\frac{T_{Air}}{T_{Smoke}-T_{Air}}=\frac{B^2}{A^2}\frac{1}{2g}\frac{T_{Air}}{\Delta T} \quad (8)$$

For minimal height of the chimney we use the equality sign.

$$\frac{h(30)}{h(-30)}=\frac{\dfrac{(30+273)}{(400)-(30)}}{\dfrac{(-30+273)}{400-(-30)}};\ h(30)=145\,m \quad (9)$$

31.(B) The velocity is constant,

$$v=\sqrt{2gh\left(\frac{\rho_{Air}}{\rho_{Smoke}}-1\right)}=\sqrt{2gh\left(\frac{T_{Smoke}}{T_{Air}}-1\right)}=\sqrt{2gh\frac{\Delta T}{T_{Air}}} \quad (10)$$

This can be seen from the equation of continuity Av = const. (ρ_{Smoke} is constant). It has a sudden jump from approximately zero velocity to this constant value when the gases enter the chimney from the furnace. In fact, since the chimney operates at minimal height this constant is equal to B, that is $v = B/A$.

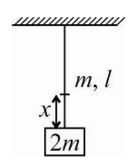
$$p_{Smoke}(z)=p(0)-(\rho_{Air}-\rho_{Smoke})gh-\rho_{Smoke}gz \quad (11)$$

Thus the pressure of smoke suddenly changes as it enters the chimney from the furnace and acquires velocity.

32.(B) We use $v_1=\sqrt{\dfrac{T_1}{\mu}}=\sqrt{\dfrac{2mg}{\mu}}=v\lambda_0$, $\qquad v_2=\sqrt{\dfrac{3mg}{\mu}}=v\lambda_2$

Comparing, we get $\lambda_2=\sqrt{\dfrac{3}{2}}\,\lambda_0$

33.(D) Let v_x be speed of pulse x distance from bottom of rope



Then $v_x=\sqrt{\left(2m+\dfrac{mx}{l}\right)\dfrac{g}{(m/l)}}=\sqrt{(2l+x)g}$

Then time taken to go from bottom to top will be

$$T=\int_0^l\frac{dx}{\sqrt{(2l+x)g}}=2\sqrt{\frac{l}{g}}(\sqrt{3}-\sqrt{2}).$$

34.(C) At instant block is in equilibrium position, its speed is maximum and compression in spring is x given by

$Kx = mg$ ... (i)

From conservation of energy

$mg(L+x)=\dfrac{1}{2}kx^2+\dfrac{1}{2}mv_{\max}^2$ ... (ii)

From (i) and (ii) we get $v_{\max}=\dfrac{3}{2}\sqrt{gL}$

35.(B) $V_{max} = \frac{3}{2}\sqrt{gL}$ *and* $\omega = \sqrt{\frac{k}{m}} = 2\sqrt{\frac{g}{L}}$ $\Rightarrow$ $A = \frac{V_{max}}{\omega} = \frac{3}{4}L$

Hence time taken t, from starts of compression till block reaches mean position is given by

$x = A\sin\omega_0 t$ where $x = \frac{L}{4}$ $\Rightarrow$ $t = \sqrt{\frac{L}{4g}}\sin^{-1}\frac{1}{3}$

Time taken by block to reach from mean position to bottom most position is $\frac{2\pi}{4\omega} = \frac{\pi}{4}\sqrt{\frac{L}{g}}$

Hence the required time $= \frac{\pi}{4}\sqrt{\frac{L}{g}} + \sqrt{\frac{L}{4g}}\sin^{-1}\frac{1}{3}$

36.(B) As, $\rho = \frac{pM}{RT}, M =$ molecular mass of the gas

AB is an isothermal process.

So, $W_{AB} = RT_A \ln\left(\frac{p_A}{p_B}\right) = RT_A \ln\left(\frac{1}{2}\right) = -\frac{p_1 M}{\rho_1}\ln(2)$

$$W_{BC} = p_B(V_C - V_B) = 2p_1\left(\frac{M}{\rho_C} - \frac{M}{\rho_B}\right) = \frac{2p_1 M}{2\rho_1} = \frac{p_1 M}{\rho_1}$$

$W_{CA} = 0,$ then $\rho =$ constant, so $V =$ constant and CA is isochoric process.

So, $W_{AB} : W_{BC} : W_{CA} :: -\ln 2 : 1 : 0$

37.(D) Heat is supplied to gas in isobaric process BC

$$Q_{BC} = W_{BC} + \Delta U_{BC} = \frac{p_1 M}{\rho_1} + C_V \Delta T = \frac{p_1 M}{\rho_1} + \frac{3}{2}R\cdot\left(\frac{p_1 M}{\rho_1 R}\right) = \frac{5p_1 M}{2\rho_1}$$

And heat is rejected in processes AB and CA

$$= Q_{AB} + Q_{CA} = \left|\left(-\frac{p_1 M}{\rho_1}\ln 2\right)\right| + \left|\left(\frac{-3p_1 M}{2\rho_1}\right)\right| = \frac{p_1 M}{\rho_1}\left(\frac{3}{2} + \ln 2\right)$$

So, ratio $= \frac{\text{Heat rejected}}{\text{Heat supplied}} = \frac{\frac{p_1 M}{\rho_1}\cdot\left(\frac{3}{2} + \ln 2\right)}{\frac{5}{2}\left(\frac{p_1 M}{\rho_1}\right)} = \frac{3 + 2\ln 2}{5} = \frac{3 + \ln 4}{5}$

38.(A) $f = \frac{C - V_R}{C + V_R} f_0$; $\frac{\Delta f}{f_0} = \frac{2V_R}{C}$

$\Rightarrow$ $C = \frac{2 \times 0.1}{600} \approx 1700\ m/s$

39.(D) $f' = \frac{C - V_R}{C + V_R} f_0$; $\frac{\Delta f}{f_0} = \frac{2V_R}{C + V_R}$

Velocity is 4 times $(A_1 v_1 = A_2 v_2)$

$\Rightarrow$ $\Delta f = 4\Delta f_0 = 2400\ Hz$

40.(A) $\frac{1}{2}\frac{dm}{dt}\left(v_2^2 - v_1^2\right) = \frac{1}{2} \times 1.5 \times 10^3 \times 0.1 \times 10^{-4} \times 0.1\left[(4 \times 0.1)^2 - (0.1)^2\right]$

$$= 0.75 \times 10^{-5}[15] = 11.25 \times 10^{-5} = 1.125 \times 10^{-4} W$$

SECTION-3 | **MULTIPLE CORRECT ANSWER TYPE**

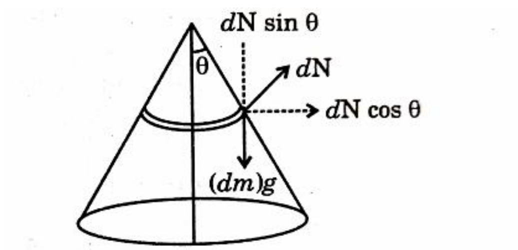
41.(AC) Rate of cooling $\frac{dT}{dt}$ is proportional to $(T-t)$, where T and t are the instantaneous temperatures of the object and environment respectively at any given time. The value of $\frac{dT}{dt}$ will therefore decreases monotonously as the temperatures T and t comes closer to each other. Similarly the rate of heat loss of the object, Q, will also decrease monotonously with time. Hence the average values $\left(\frac{dT}{dt}\right)$ and (Q) will both decrease with time. Hence $\alpha_A > \alpha_B$ and $\beta_A > \beta_B$.

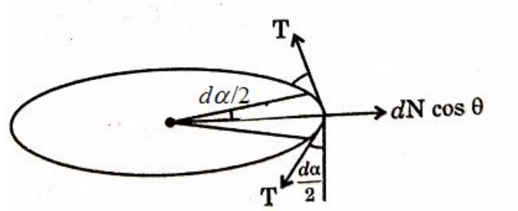
42.(ABC) $dN \sin\theta = (dm)g$

So, $N \sin\theta = mg$

$$\frac{N}{\sqrt{2}} = mg$$

$$N = \sqrt{2}mg \qquad \textbf{....(i)}$$





$$2T \sin\left(\frac{d\alpha}{2}\right) = dN \cos\theta$$

$$2T\left(\frac{d\alpha}{2}\right) = dN \cos\theta \qquad \text{[For small angle } \sin\alpha = \alpha \text{]}$$

$$T(2\pi) = \sqrt{2}mg \times \frac{1}{\sqrt{2}} \qquad \textbf{....(ii)}$$

$$T = \frac{mg}{(2\pi)}$$

$$\frac{T}{A} = Y\frac{\Delta l}{l} \qquad \frac{mg}{2\pi A} = \frac{Y\Delta l}{2\pi R}$$

$$\Delta l = \frac{mgR}{Ay}$$

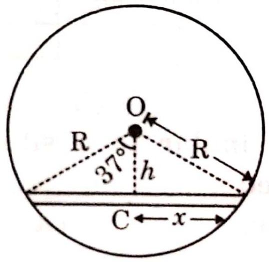
43.(AC) $Y = \frac{\text{Stress}}{\text{Strain}}$ for same strain $(\text{stress})_A > (\text{stress})_B$

$Y_A > Y_B$

Breaking point of B > Breaking point of A

So, A is less ductile.

44.(BD) Consider squirrel at distance x from midpoint C of platform.



F.B.D. of squirrel :

$f = ma$ **. . . .(i)**

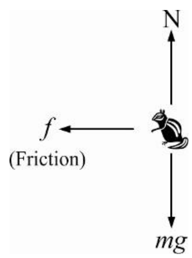
$N = mg$ **. . . .(ii)**

F.B.D. of platform + Wheel cage

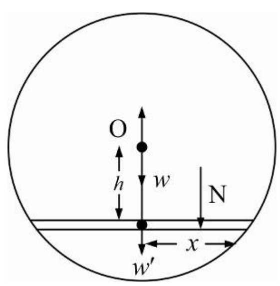
Platform + wheel cage remain at equilibrium only if $\tau_{\text{pivot}} = 0$, which is possible according to the following F.B.D. only.

$\Rightarrow$ $fh = Nx$

$\Rightarrow$ $mah = mg\,x$ [From equations (i) and (ii)]



$a = \dfrac{gx}{h}$ towards C (restoring acceleration)



$\therefore$ Squirrel performs SHM about mean position C, angular frequency $\omega = \sqrt{\dfrac{g}{h}}$ and amplitude

$A = \dfrac{3R}{5}$ (Half the platform's length)

In SHM, velocity at extreme should be zero.

Squirrel is at rest at one end of the platform at $t = 0$.

Maximum velocity $= A\omega = \dfrac{3R}{5}\sqrt{\dfrac{g}{h}}$

$$v_{max} = \frac{3R}{5}\sqrt{\frac{g}{\frac{4R}{5}}}$$

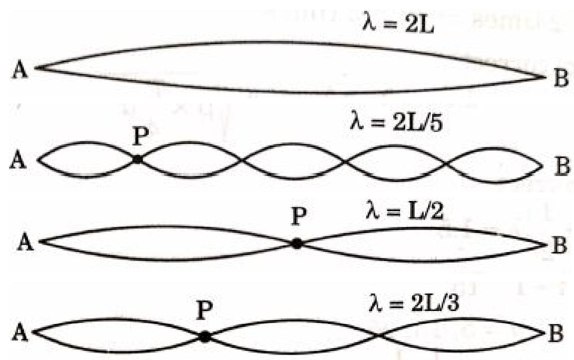
$$v_{max} = \sqrt{\frac{9Rg}{25} \cdot \frac{5}{4}}$$

$$v_{max} = \sqrt{\frac{9Rg}{20}}$$

Maximum acceleration $= \omega^2 A$

$$a_{max} = \frac{gA}{h} = \frac{g\frac{3R}{5}}{\frac{4R}{5}} = \frac{3g}{4}$$

45.(ABD)



46.(AD) At $z = -0.01$, $x = max$ $\therefore$ Upper end-open

$z = 0.99\,m$, $x = min$ lower end-closed

$$\frac{2\pi}{0.8} = \frac{2\pi}{\lambda}$$

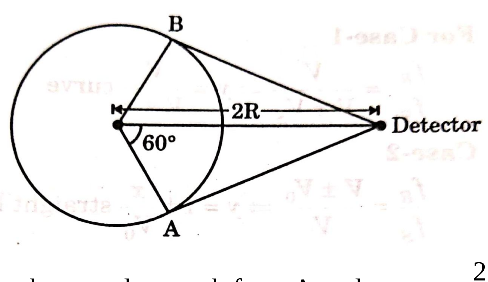
$\Rightarrow$ $\lambda = 0.8\,m$

$l_{eff.} = 0.99 + 0.01 = 1\,m$

$\frac{l}{\lambda} = \frac{5}{4},\ l = 5\left(\frac{\lambda}{4}\right)$ five half loops

$\Rightarrow$ Seconds overtone

47.(ABCD) Sound emitted by source at A will result in maximum frequency while sound emitted by source at B will result in minimum frequency.



Time taken by sound to reach from A to detector $= \frac{2R\sin 60°}{330}$

Angle traveled by source by this $\frac{330\pi}{6\sqrt{3}R} \times \frac{2R\sin 60°}{330} = \frac{\pi}{6}$

$$f_{max} = \left(\frac{330}{330 - \frac{330\pi}{6\sqrt{3}}}\right) f;\quad f_{min} = \left(\frac{330}{330 + \frac{330\pi}{6\sqrt{3}}}\right) f$$

48.(AD) $v_1 = \sqrt{2gh}$ $\quad v_2 = \sqrt{2g(h+H)}$

$x = v_1 t_1 = v_2 t_2$ and $y = h + \frac{1}{2} g t_1^2$

and $y = h + H + \frac{1}{2} g t_2^2$

The point of intersection will remain at the same place if the level of water in the vessel does not change. That means $Q = A_1 v_1 + A_2 v_2$

or $Q = a(v_1 + v_2)$, $\quad v_1 + v_2 = \frac{Q}{a}$

$\Rightarrow \quad \sqrt{2gh} + \sqrt{2g(h+H)} = \frac{Q}{a}$, $\sqrt{2g(h+H)} = \frac{Q}{a} - \sqrt{2gh}$

On squaring both sides, we get : $2gh + 2gH = \frac{Q^2}{a^2} + 2gh - 2\frac{Q}{a}\sqrt{2gh}$

$2\frac{Q}{a}\sqrt{2gh} = \frac{Q^2}{a^2} - 2gH$, $\quad 2\left(\frac{140}{0.2}\right)\sqrt{2 \times 980 \times h} = \frac{140^2}{(0.2)^2} - 2 \times 980 \times 50$

$\Rightarrow \quad h = 40$ cm $\quad \therefore \quad v_1 = \sqrt{2 \times 980 \times h} = 280 \text{ cms}^{-1}$

and $\quad v_2 = \sqrt{2 \times 980(40+50)} = 420 \text{ cms}^{-1}$

$t_1 = \frac{v_2}{v_1} t_2 = \frac{420}{280} t_2 = \frac{3}{2} t_2$

$\therefore \quad y = 40 + \frac{1}{2} \times 980 \times \frac{9}{4} t_2^2 = 40 + 50 + \frac{1}{2} \times 980 t_2^2$

$\frac{1}{2} \times 980 \left(\frac{5}{4} t_2^2\right) = 50 \quad \Rightarrow \quad t_2 = \left(\frac{4}{49}\right) = \frac{2}{7}$s

$\therefore \quad t_1 = \frac{3}{7}$s $\quad \therefore \quad x = v_1 t_1 = (280)\left(\frac{3}{7}\right) = 120$ cm

and $\quad y = h + \frac{1}{2} g t_1^2 \quad = 40 + \frac{1}{2}(980)\left(\frac{9}{49}\right) = 130$ cm

49.(ABCD) $f_0 = \frac{1}{2L}\sqrt{\frac{F}{\mu}} = \frac{1}{2L}\sqrt{\frac{F}{\rho A}}$

Where, F = tension

$\mu =$ mass per unit length

Stress $= \frac{F}{A} = Y$ (Strain) $= Y\left(\frac{\Delta L}{L}\right) = Y\left(\frac{L\alpha\theta}{L}\right) = Y\alpha\theta$

$\therefore \quad f_0 = \frac{1}{2L}\sqrt{\frac{Y\alpha\theta}{\rho}} \quad \therefore \quad f_0 \propto \frac{1}{L} \propto \sqrt{Y} \propto \sqrt{\theta} \propto \sqrt{\frac{\alpha}{\rho}}$

50.(BCD) Due to the pseudo force on block (considered external) its mean position will shift to a distance mg/K above natural length of spring as net force now is mg in upward direction. So total distance of block from new mean position is 2mg/K which will be the amplitude of oscillations hence option (C) is correct. During oscillations spring will pass through the natural length hence option (D) is correct As block is oscillating under spring force and other constant forces which do not affect the SHM frequency hence option (B) is correct.

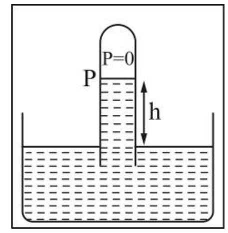
51.(ABCD) By Dulong-petit law, molar specific heat of crystalline solids is $3R$

52.(BD) $r' = r(1 + \alpha\Delta T)$

$d' = d(1 + \alpha\Delta T)$

The total length will also increase $\ell' = \ell(1 + \alpha\Delta T)$

As length and radius increases by same factor, their ratio (angle) remains same.



53.(BCD) When stationary $\rho gh = P$

When accelerating upwards $\rho(g+a)h_1 = P \quad \Rightarrow \quad h_1 < h$

When accelerating downwards $\rho(g-a)h_2 = P \quad \Rightarrow \quad h_2 > h$

When moving with constant velocity $\rho gh_3 = P \quad \Rightarrow \quad h_3 = h$

54.(BD) KE for an ideal gas $= \frac{f}{2}nRT = \frac{5}{2}nRT$

Also, KE of gas w.r.t. ground = KE of COM w.r.t. ground + KE of gas w.r.t. COM of system

$$= \frac{1}{2}mv^2 + \frac{5}{2}nRT$$

55.(AC) When the two waves superpose then the maximum amplitude will be 4 + 3 = 7 and minimum will be 4 – 3 =1 hence the ratio of maximum to minimum intensity will be 49 and as the frequencies of the two waves are 200 Hz and 202 Hz the beat frequency will be 2Hz.

56.(AC) Volume decreases during melting of ice.

$\therefore$ Positive work is done on the ice-water system by the atmosphere

Heat is absorbed by ice-water system. So, internal energy increases.

57.(ABC) $\omega' = \sqrt{\frac{k}{m_1}}; \quad v_1 = \sqrt{\frac{k}{m_1+m_2}}\, x_0$

$$\omega' A' = \sqrt{\frac{k}{m_1+m_2}}\, x_0 \qquad ; \qquad A' = \sqrt{\frac{m_1}{m_1+m_2}}\, x_0$$

58.(AC) Because 90% of the volume of the ice is inside water, the relative density of ice must be 0.9. The ice block is pushed out of the water as the kerosene is poured. This is due to increased pressure at the bottom of the ice block. After the ice block gets completely submerged, there is no change in pressure difference at the bottom surface and the top surface. The ice block does not move after this.

Let A = area of cross section of the ice cylinder

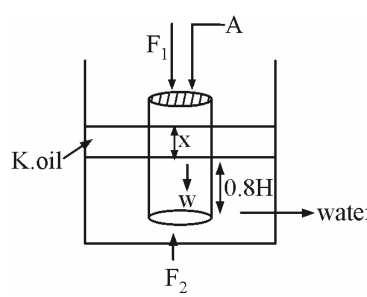
H = height of the cylinder

F_1 = force on top surface due to atmospheric pressure (P_0)

F_2 = force on bottom surface due to water pressure

W = weight of the cylinder

For equilibrium



$$F_2 - F_1 = W \qquad \textbf{....... (i)}$$

$$\Rightarrow \quad [P_0 + \rho_k gx + \rho_w g(0.8H)]A - P_0 A = AH\rho_{ice} g$$

$$\frac{\rho_k}{\rho_w}x + 0.8H = \frac{\rho_{ice}}{\rho_w}H$$

$$0.8x + 0.8H = 0.9H \quad \Rightarrow \quad x = \frac{H}{8}$$

[Alternative way of writing equation (i) is thinking in terms of Archimedes principle]

Buoyancy $= W$

Weight of water displaced + weight of kerosene displaced $= W$

$$0.8HA\rho_w g + xA\rho_k g = HA\rho_{ice} g$$

59.(BC)

60.(ABD) Work done = area under the curve,

$$W_1 = \left(P + \frac{P}{2}\right)(2V - V) = \frac{3}{2}PV$$

Work done under isotherm process,

$$W_2 = RT \times 2.3026\ log\left(\frac{2V}{V}\right) = 0.693\,RT = 0.693\,PV$$

$\therefore$ $W_1 > W_2$, i.e., option (A) is correct.

Let P_0 and V_0 be the intercepts on the P and V axes. Now the equation of straight line would be

$$P = -\frac{P_0}{V_0} \times V + P_0 \qquad (\because\ y = mx + c)$$

Further $PV = RT$ or $P = \frac{RT}{V}$

(This represents parabola, so option (B) is also correct.)

SECTION-4	COLUMN MATCHING TYPE

61. **[A-r]** **[B-p]** **[C-s]** **[D-q]**

(A) $$\frac{\Delta Q}{\Delta t} = \frac{k_0\left[\pi(2R)^2 - \pi R^2\right]}{3R}(T_1 - T_2) = \pi R k_0 (T_1 - T_2)$$

(B) $$\frac{\Delta Q}{\Delta t} = \frac{4\pi k R_1 R_2}{(R_2 - R_1)}(T_1 - T_2) = \frac{4\pi k_0 3R \cdot R}{(3R - R)}(T_1 - T_2) = 6\pi k_0 R (T_1 - T_2)$$

(C) $$\frac{\Delta Q}{\Delta t} = \frac{2\pi k_0 l}{ln\left(\frac{R_2}{R_1}\right)}(T_1 - T_2) = \frac{4\pi k_0 R}{ln\,2}(T_1 - T_2)$$

(D) $$\frac{\Delta Q}{\Delta t} = -\pi R^2 k \cdot \frac{dT}{dx} = -\pi R^2 k_0 \left(1 + \frac{x}{3R}\right)\frac{dT}{dx}$$

$$\therefore \quad \frac{\Delta Q}{\Delta t} = \int_0^{3R} \frac{dx}{\left(1 + \frac{x}{3R}\right)} = -\pi R^2 k_0 \int_{T_1}^{T_2} dt$$

$$\frac{\Delta Q}{\Delta t} = \left.\frac{ln\left(1 + \frac{x}{3R}\right)}{\left(\frac{R}{3}\right)}\right|_0^{3R} = \pi R^2 k_0 (T_1 - T_2)$$

$$\therefore \quad \frac{\Delta Q}{\Delta t} = \frac{\pi k_0 R (T_1 - T_2)}{3\,ln\,2}$$

62. **[A-r]** **[B-s]** **[C-p]** **[D-q]**

(A, B) Heat, ΔH added to the gas along the straight line path AC.

Work ΔW_{AC} done by the gas along the straight line path $AC = \text{Area}\left(\int_A^C pdv\right)$ under the path $\Delta W_{AC} = P_A V_A + \frac{1}{2} P_A V_A = \frac{3}{2} P_A V_A$

Increase of internal energy ΔU_{AC} is

$$\Delta U_{AC} = \frac{3}{2} nR (T_C - T_A) = \frac{3}{2}(P_C V_C - P_A V_A) = \frac{9}{2} P_A V_A$$

Hence heat added,

$$\Delta H_{AC} = \Delta W_{AC} + \Delta U_{AC} = \frac{3}{2}P_A V_A + \frac{9}{2}P_A V_A = 6P_A V_A$$

(C) Work done

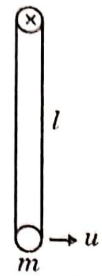
$$\Delta W_{ABC} = \Delta W_{AB} + \Delta W_{BC} = 0 + P_C\left(V_C - V_A\right) = 2P_A V_A$$

(D) Heat added to the gas along ABC,

$$\Delta H_{ABC} = \Delta U + \Delta W = \frac{9}{2}P_A V_A + 2P_A V_A$$

63. **[A-p, q, s, t]** **[B-p, q, s]** **[C-p, q, r, s]** **[D-p, q, s, t]**

For (A) :

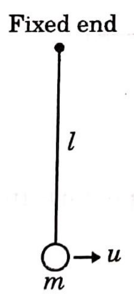


(p) $T = 2\pi\sqrt{\frac{l}{g}}$ simple pendulum (only for small oscillations)

(q) $u = \sqrt{4.5\,gl} > \sqrt{4\,gl}$ so, it will complete circular motion and at highest point $mg >$ centrifugal force. So, rod will be having compression force.

(t) For $u = \sqrt{3gl}$, the rod will perform oscillations of amplitude $\frac{2\pi}{3}$

For (B) :



(p) $T = 2\pi\sqrt{\frac{l}{g}}$ simple pendulum (only for small oscillations)

(q) $u = \sqrt{4.5\,gl} < \sqrt{5\,gl}$ so string will slack

(t) String will slack, so angular amplitude of $\frac{2\pi}{3}$ is not possible.

For (C) : $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m}{\left(\frac{mg}{l}\right)}} = 2\pi\sqrt{\frac{l}{g}}$

(q) $u = \sqrt{4.5\,gl}$, block will rise above spring natural length position during oscillations, so, spring will be compressed.

(t) No angular oscillations.

For (D) :

(p) $T = 2\pi\sqrt{\dfrac{I}{mgd}} = 2\pi\sqrt{\dfrac{\left(\dfrac{ml^2}{12} + mr^2\right)}{mgr}}$ (for small oscillations)

So, for $T = 2\pi\sqrt{\dfrac{l}{g}} \Rightarrow r = \dfrac{l}{2} - \dfrac{l}{\sqrt{6}}$

(q) $u = \sqrt{4.5\,gl}$, rod may complete circular motion and on top, it is possible that $g >$ centrifugal force, so, rod may experience compression force.

(r) For $u = \sqrt{3gr}$, angular amplitude of $\dfrac{2\pi}{3}$ is possible

64. **[A-r]** **[B-p]** **[C-p]** **[D-r]**

$PV^{\gamma} =$constant, $V^{(\gamma-1)}T$ = constant,

$P^{(1-\gamma)}T^{\gamma} =$constant

$P \propto \dfrac{1}{V^{\gamma}}, P \propto T^{\gamma/(\gamma-1)}, V \propto T^{1(\gamma-1)}$

$P \propto T^{\gamma/(\gamma-1)} \propto (I.E.)^{\gamma/(\gamma-1)}$

65. **[A-q]** **[B-p]** **[C-r]** **[D-s]**

(A) $\lambda = L$ as $x = 0$ and $x = L$ are nodes

$\therefore$ Terms should be either $\sin\dfrac{\pi x}{L}$ or $\sin\dfrac{2\pi x}{L}$ as $k = \dfrac{2\pi}{L} = \dfrac{2\pi}{\lambda} \Rightarrow \lambda = L$

$\therefore$ (Q) is suitable for (A).

(B) $\lambda = 2L$ as $x = 0$ and $x = L$ are nodes $\therefore$ Terms should be either $\sin\dfrac{\pi x}{L}$ or $\sin\dfrac{2\pi x}{L}$

As $k = \dfrac{\pi}{L} = \dfrac{2\pi}{\lambda} \Rightarrow \lambda = 2L$ $\therefore$ (P) is suitable for (B)

(C) $\lambda = 4L$ as $x = 0$ is node $\therefore$ Terms should be either $\cos\dfrac{\pi x}{2L}$ *or* $\cos\dfrac{2\pi x}{L}$

As $k = \dfrac{2\pi}{\lambda} = \dfrac{\pi}{2L} \Rightarrow 4L$

$\therefore$ (R) is suitable for (C)

(D) $\lambda = 2L$ as $x = 0$ is node

$\therefore$ Terms should be either $\cos\dfrac{\pi x}{L}$ *or* $\cos\dfrac{\pi x}{2L}$

As $k = \dfrac{\pi}{L} = \dfrac{2\pi}{\lambda} = \lambda = 2L$ $\therefore$ (s) is suitable for (D)

66. **[A-q]** **[B-r]** **[C-p]** **[D-s]**

(P) Pressure above surface $\times$ Area

$\pi R^2 \times \rho g h$

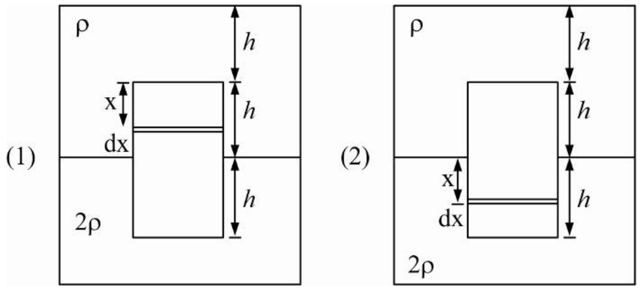
(B) Pressure below bottom $\times$ Area $= (2\rho g h + 2\rho g h)\pi R^2 = 4\pi R^2 \rho g h$

(C) $F_V = \int_0^h \rho g(h + x)\,dx \times 2R = \rho R g 3h^2$

$$F_r = \int_0^h (2\rho gh) + (2\rho gx))\ 2Rdx$$

$$= 4\rho gh \frac{3h^2}{2} \Rightarrow 6\rho Rgh^2$$

Total = $9\rho Rgh^2$



(D) $F_{Buoyancy} = \rho\alpha pR^2hg + 2\rho\pi R^2 hg$

$$= 3\rho\pi R^2 gh$$

67. **[A-q]** **[B-p]** **[C-s]** **[D-r]** Analyse using circular motion analogy.

68. **[A-q]** **[B-s]** **[C-r]** **[D-r]** Use : $\lambda' = \frac{V_{relative}}{f'}$

69. **[A-p]** **[B-r]** **[C-r]**

(A) Before throwing iron into water, iron had displaced water equal to its weight, while displaced water equal to its weight, while after throwing it would displace water equal to its volume.

(B) In both cases, log would displace water equal to its weight.

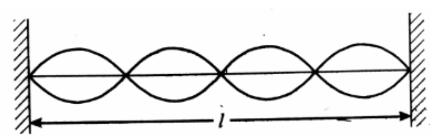
(C) In both cases, water would displace water of swimming pool equal to its weight.

70. **[A-p, r, s]** **[B-p, r, s]** **[C-p]** **[D-p, q, r, s]**

SECTION-5	NUMERICAL TYPE

71.(0.87) For a string vibrating in its nth overtone [$(n+1)^{th}$ harmonic]

$$y = 2A\sin\left(\frac{(n+1)\pi x}{L}\right)\cos\omega t$$



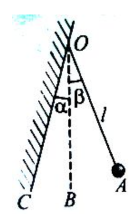
For $x = \frac{l}{3}$, $2A = a$ and $n = 3$; $y = \left[a\sin\left(\frac{4\pi}{l}.\frac{l}{3}\right)\right]\cos\omega t$

$$= a.\sin\frac{4\pi}{3}\cos\omega t = -a.\left(\frac{\sqrt{3}}{2}\right)\cos\omega t$$ i.e. at $x = \frac{l}{3}$; the amplitude is $\frac{\sqrt{3}a}{2}$.

72.(12) Relative to liquid, the velocity of sphere is $2v_0$ (upwards).

$\therefore$ Viscous force on sphere = $6\pi\eta r(2V_0)$ (downwards) $= 12\pi\eta rv_0$ (downwards)

73.(1.33) The time period of free oscillation of pendulum $T = 2\pi\sqrt{\frac{l}{g}} = 2s$



Time taken by bob to go from extreme position A to mean position B is $= \frac{T}{4}$

Time taken by bob to move from mean position B to position C (where its angular displacement α is half the angular amplitude β) is found from equation $\alpha = \beta\sin\left(\frac{2\pi}{T}t\right)$

Solving we get $t = \frac{T}{12}$ $\Rightarrow$ Total time period of oscillation is $= 2\left[\frac{T}{4} + \frac{T}{12}\right] = \frac{2}{3}T = \frac{4}{3}s$

74.(– 50) As, $\frac{C-0}{100-0} = \frac{N-100}{400-100}$ $\Rightarrow$ $\frac{x}{100} = \frac{x-100}{300}$

$\Rightarrow x = \frac{x-100}{3}$ $\Rightarrow x = -50°C$

75.(2) As temperature changes by ΔT , new length l' and moment of inertia I' are given as :

$l' = l(1+\alpha\Delta T)$ $\qquad I' = I(1+2\alpha\Delta T)$

Thus new time period is

$$T' = 2\pi\sqrt{\frac{I'}{mgl'}} \qquad = 2\pi\sqrt{\frac{I}{mgl}}\frac{(1+2\alpha\Delta T)^{1/2}}{(1+\alpha\Delta T)^{1/2}}$$

As $\alpha\Delta T << 1$, using binomial approximate we get

$T' = 2\pi\sqrt{\frac{I}{mgl}}\left(1+\frac{1}{2}\alpha\Delta T\right)$ change in period is $T' - T = 2\pi\sqrt{\frac{I}{mgl}}\left(\frac{1}{2}\alpha\Delta T\right) = \pi\alpha\Delta T\sqrt{\frac{I}{mgl}}$

76.(3) Given that from graph

$$\frac{1}{2}mV_m^2 = 15\times 10^{-3} \qquad \Rightarrow \qquad V_m = \sqrt{0.150}\ m/s$$

$$A\omega = \sqrt{0.150}\ m/s \qquad \Rightarrow \qquad L\theta_m,\sqrt{\frac{g}{L}} = \sqrt{0.150}\ m/s$$

$$\Rightarrow \sqrt{gL} = \frac{\sqrt{0.150}}{100\times 10^{-3}} \qquad \Rightarrow \qquad L = \frac{0.150}{0.1} = 1.5m$$

77.(5.28) On increasing temperature of ball by 100°C (from 20°C to 120 °C), the thermal expansion in its volume can be given as

$$\Delta V = \gamma_{st}V\Delta T \quad = 3\alpha_{st}V\Delta T \qquad \text{... (i)}$$

Here it is given that no change is volume is allowed. This implies that the volume increment by thermal expansion is compressed elastically by external pressure. Thus elastic compression in the sphere must be equal to that given in equation (i). Bulk modulus of a material is defined as

$$B = \frac{\text{increase in pressure}}{\text{volume strain}} = \frac{\Delta P}{\Delta V/V}; \qquad \Delta P = B\times\frac{\Delta V}{V} = B(3\alpha_{st}\Delta T)$$

$$= 1.6\times 10^{11}\times 3\times 1.1\times 10^{-5}\times 100 \qquad = 5.28\times 10^{8}\,Pa$$

Thus this must be the excess pressure inside the ball at $120°C$ to keep its volume constant during heating.

78.(0.6) Let volume of block $= V_1$

Volume of concrete $= V_2$

$\therefore$ Displaced volume of water $= (V_1+V_2)$

Now, weight of the combination = Buoyant force

$\therefore \quad 0.5\times V_1 g + 0.25\times V_2 g = 1\times(V_1+V_2)g$

$\therefore \quad \frac{V_1}{V_2} = \frac{3}{1} \qquad \therefore \qquad \frac{m_1}{m_2} = \frac{0.5\times V_1}{0.25\times V_2} = 3/5$

79.(6) For equilibrium of block in frame of container, we use

$$F_B = T + mg_{eff}$$

$$T = F_B - mg_{eff} = V\rho g_{eff} - mg_{eff} = (1000)\times 1.2\times 1500 - (1000)\times 0.8\times 1500$$

$$= 6\times 10^5\,dyne = 6N$$

80.(0.15) Velocity gradient in oil layer :

$$\frac{dv}{dx} = \frac{v}{x} = \frac{0.5}{0.15\times 10^{-2}} = \frac{1000}{3}sec^{-1}$$

For steady speed of block :

Viscous force on block $F = mg\sin\theta$

$$\eta A\frac{dV}{dx}=(13)(10)\left(\frac{5}{13}\right)$$

$$\eta=\frac{50}{1\times(1000/3)}=\frac{15}{100}=0.15$$

81.(6) Mass of rod $m=A(2l)(0.75P_{\omega})=1.5\,Al\rho_{\omega}$

Buoyancy on rod $F_B=A(2l-x)\,\rho_{\omega}g$

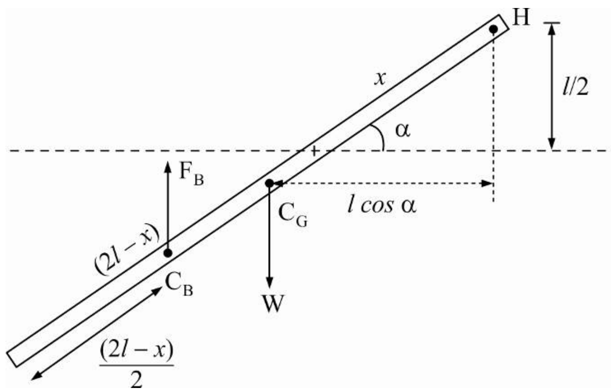
For equilibrium, not torque on rod must be zero about any point :

$$\tau_H=mg\times l\cos\alpha-F_B\times\left[2l-\frac{2l-x}{2}\right]\cos\alpha=0$$

$$=1.5\,Al\rho_{\omega}g\,.l-A(2l-x)\,\rho_{\omega}g\;.\left(\frac{2l+x}{2}\right)=0$$

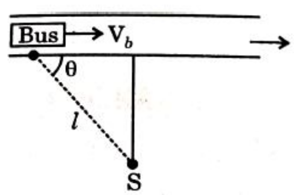
$\Rightarrow\ 1.5l^2-2l^2+\dfrac{x^2}{2}=0$ From figure

$$\frac{x^2}{2}=\frac{l^2}{2}\ \Rightarrow\ x=\ell\ \Rightarrow\ \sin\alpha=\frac{l}{2x}=\frac{1}{2}\ \Rightarrow\ \alpha=30^\circ$$



82.(144)

$$\frac{V_{sound}}{V_{sound}-V_{bus}\cos\theta}(1000)=1500$$



$$\frac{1}{1-\frac{2}{\sqrt{3}}\cos\theta}=\frac{3}{2}$$

$$\theta=30^\circ$$

$$\frac{l}{V_{sound}}+\frac{l\sin\theta}{V_{bullet}}=\frac{l\cos\theta}{V_{bus}}$$

$$\frac{1}{2V_{bullet}}=\frac{1}{V_{sound}}\left[\frac{\sqrt{3}V_{sound}}{2V_{bus}}-1\right]$$

$$\frac{1}{2V_{bullet}}=\frac{1}{V_{sound}}\left[\frac{\sqrt{3}}{2}\cdot\frac{3\sqrt{3}}{2}-1\right]=\frac{5}{4V_{sound}}$$

$$V_{bullet}=\frac{2}{5}V_{sound}=\frac{2}{5}(360)=4(36)=144=(12)^2$$

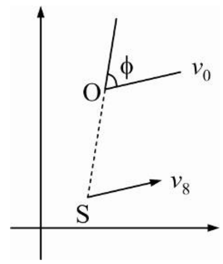
83.(1) Phase difference between S_1 and S_2 $= \dfrac{2\pi}{\lambda} \times \Delta x = \dfrac{2\pi}{\frac{340}{170}} \times 1 = \pi$

$$I_{S_1} = I_{S_2} = I_{S3} = 1$$

Intensity due to all three source at point P is = I

84.(7) $\overrightarrow{SO} = 3\hat{i} + 4\hat{j}$

Component of velocity along SO



$$v_s = 15\left(2\hat{i} + \hat{j}\right) . \left(\frac{3\hat{i} + 4\hat{j}}{5}\right) = 30$$

$$v_0 = \left(5\hat{i} + 15\hat{j}\right) . \left(\frac{3\hat{i} + 4\hat{j}}{5}\right) = 15$$

$$f = \frac{330 - 15}{330 - 30} \times 100 = 105\, Hz \quad \Rightarrow \quad 15x = 105 \Rightarrow$$

85.(4) In first case $\dfrac{l + 2l + \ldots + nl}{k_{eq} A} = \dfrac{l}{kA} + \dfrac{2l}{2kA} + \dfrac{3l}{3kA} + \ldots + \dfrac{nl}{nkA}$

$$k_{eq_1} = \frac{(n+1)k}{2}$$

In second case

$$k_{eq_2} = \frac{\sum k_i A_i}{\sum A_i} = \frac{kA' + (2k)(2A') + \ldots + (nk)(nA')}{A' + 2A' + \ldots + nA'} = \frac{(2n+1)k}{2}$$

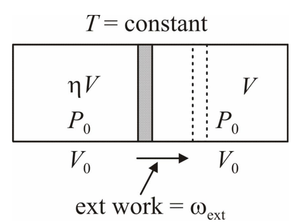
$$\frac{(2n+1)k}{3} = \frac{6}{5}\frac{(n+1)k}{2}$$

$$9n + 9 = 10n + 5$$

$$n = 4$$

SECTION-6	SUBJECTIVE TYPE

86. $\boxed{P_0 V_0 \ln\left[\dfrac{(\eta+1)^2}{4\eta}\right]}$



After displacement of piston $V + \eta V = 2V_0$ $\qquad V = \dfrac{2V_0}{\eta + 1}$

By energy conservation

$$\left|\omega_{\text{by left gas}}\right| + \left|\omega_{\text{ext}}\right| = \left|\omega_{\text{on right gas}}\right|$$

$$\omega_{\text{ext}} = \left|\omega_{\text{on right gas}}\right| - \left|\omega_{\text{by left gas}}\right| = P_0 V_0 \ln\left(\frac{V_0}{V}\right) - P_0 V_0 \ln\left(\frac{\eta V}{V_0}\right)$$

$$= P_0 V_0 \ln\left[\frac{(\eta+1)^2}{4\eta}\right]$$

87. $\boxed{2\pi\left(\dfrac{2\sqrt{2m}\, p^{3/2}}{q^2}\right)}$

As potential energy of particle is given as $U_{(x)} = \dfrac{p}{x^2} - \dfrac{q}{x}$

The force on particle can be given as $F=\left|\frac{dU}{dx}\right|=\left|-\frac{2p}{x^3}+\frac{q}{x^2}\right|$

The equilibrium position of particle can be given for $F=0,$ as $-\frac{2p}{x^3}+\frac{q}{x^2}=0$

or $\quad x=\frac{2p}{q}$

Thus at $x={}^{2p}\!/_{q}$ position, particle is in equilibrium. If we find restoring force on particle if it is displaced slightly by a distance z along + x direction then it is given as

$$F_R=-\left[-\frac{2p}{({}^{2p}\!/_{q}+z)^3}+\frac{q}{({}^{2p}\!/_{q}+z)^2}\right]$$ [–ve sign for restoring nature]

For small z we have

$$F_R=-\left[-\frac{2p}{({}^{2p}\!/_{q})^3}\left(1+\frac{qz}{2p}\right)^{-3}+\frac{q}{({}^{2p}\!/_{q})^2}\left(1+\frac{qz}{2p}\right)^{-2}\right] \text{ or } F_R=-\left[-\frac{q^3}{4p^2}\left(1-\frac{3qz}{2p}\right)+\frac{q^3}{4p^2}\left(1-\frac{bz}{p}\right)\right]$$

$$F_R=-\left[\frac{3q}{2p}-\frac{q}{p}\right]\frac{q^3}{4p^2}z \qquad F_R=-\left(\frac{q^4}{8p^3}\right)z$$

If a is the acceleration of particle, we have $a=-\left(\frac{q^4}{8mp^3}\right)z$

Comparing with basic differential equation of SHM, we get the angular frequency of its oscillations as

$$\omega=\sqrt{\frac{q^4}{8mp^3}}=\frac{q^2}{2\sqrt{2m}\,p^{3/2}}$$

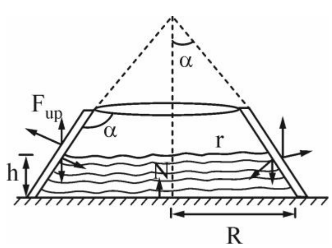
The time period of its oscillations is $T=\frac{2\pi}{\omega}=2\pi\left(\frac{2\sqrt{2m}\,p^{3/2}}{q^2}\right)$

88. $\boxed{\dfrac{W}{\pi h^2 g\tan\alpha\left(R-\frac{h}{3}\tan\alpha\right)}}$

Normal reaction at bottom due to liquid pressure $N=h\rho g.\,\pi R^2$
For equilibrium of liquid $N = mg + F_{up}$
Upward force on vessel walls is

$F_{up}=h\rho g\pi R^2-mg=W$ (wt of vessel)

$h\rho g\pi R^2-V\rho g=W \qquad \Rightarrow \qquad \rho=\frac{W}{hg\pi R^2-Vg}$



Volume of liquid:

$$V=\frac{1}{3}\pi R^3\cot\alpha-\frac{1}{3}\pi r^3\cot\alpha, \text{ where } r=R-h\tan\alpha$$

Solving, we get $\rho=\dfrac{W}{\pi h^2 g\tan\alpha\left(R-\frac{h}{3}\tan\alpha\right)}$

89. $\boxed{k = 1.713\ W/m°C,\ \dot{m} = 1.74 \times 10^{-8}\ kg/s}$

It is given that the system is in steady state. This means that any part of rod is not absorbing any heat. So heat absorbed by the end of the rod which is in vacuum chamber by radiation is fully conducted to the ice bath through the rod. Thus we have

Rate of heat conduction through the rod = Rate of heat absorption by radiation from the vacuum chamber

or $\dfrac{kA(T_B - T_A)}{L} = \sigma A(T_{vc}^4 - T_B^4)$

or $\dfrac{k(17°C - 0°C)}{0.5} = 5.67 \times 10^{-8}[(300)^4 - (290)^4]$

or $k = \dfrac{5.67 \times 10^{-8}[(300)^4 - (290)^4] \times 0.5}{17} = 1.713\ W/m°C$

Using this value of k we can find the rate of heat obtained by the ice bath as

$$\frac{dQ}{dt} = \frac{kA(T_A - T_B)}{l} = \frac{1.713 \times 1 \times 10^{-4} \times 17}{0.5} = 5.82 \times 10^{-3}\ J/s$$

This heat is used to melt the ice in ice bath. If m mass of ice is being melted per second, then we have

$\dfrac{dQ}{dt} = mL$ or $5.82 \times 10^{-3} = m \times 3.35 \times 10^5$ or $m = 1.74 \times 10^{-8}$ kg/s.

90. $\boxed{\dfrac{\sqrt{3}\ \Delta\theta}{2(\alpha_2 - \alpha_1)}}$

$$\sin\left(30° + \frac{\Delta\theta}{2}\right) = \frac{\frac{\ell}{2}(1 + \alpha_2 \Delta T)}{\ell(1 + \alpha_1 \Delta T)}$$

$$\sin 30° \cos\frac{\Delta\theta}{2} + \cos 30° \sin\frac{\Delta\theta}{2} = \frac{1}{2}(1 + \alpha_2 \Delta T)\ (1 + \alpha_1 \Delta T)^{-1}$$

$\because$ $\Delta\theta$ is very small : $\cos\dfrac{\Delta\theta}{2} \approx 1,\ \sin\dfrac{\Delta\theta}{2} = \dfrac{\Delta\theta}{2}$

$\Rightarrow$ $\dfrac{1}{2}(1) + \dfrac{\sqrt{3}}{2}\left(\dfrac{\Delta\theta}{2}\right) = \dfrac{1}{2}\left[1 + (\alpha_2 - \alpha_1)\ \Delta T\right]$ $\Rightarrow$ $\Delta T = \dfrac{\sqrt{3}\ \Delta\theta}{2(\alpha_2 - \alpha_1)}$

91. $\boxed{\dfrac{1}{R}[P_0 \Delta S + (m_1 + m_2)_g]}$

For equilibrium of piston, if T is tension in string and P_g is gas pressure

For m_1: $P_0 s_1 + m_1 g + T = P_g S_1$ . . .(i) ; For m_2: $P_g S_2 + m_2 g = T + P_0 S_2$ . . . (ii)

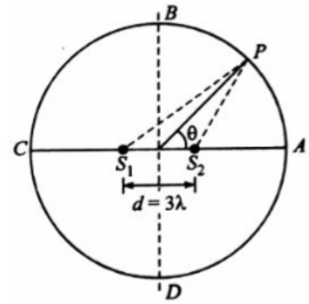
(i) + (ii): $P_0(s_1 - s_2) + (m_1 + m_2)g = P_g(s_1 - s_2)$

$P_g = P_0 + \dfrac{(m_1 + m_2)g}{\Delta s}$ $\Rightarrow$ P_g remains same

By gas law, at temperature T_1 : $P_g V = RT_1$: At temp $T_2 (T_2 > T_1)$: $P_g(V + \ell \Delta s) = RT_2$

$P_g \Delta s \ell = R(T_2 - T_1)$ $\Rightarrow$ $T_2 - T_1 = \dfrac{P_g \Delta s \ell}{R}$ $= \dfrac{1}{R}[P_0 \Delta S + (m_1 + m_2)_g]$

92. $\boxed{\mathbf{0, \cos^{-1}\left(\frac{2}{3}\right), \cos^{-1}\left(\frac{1}{3}\right), \frac{\pi}{2}} \textbf{ in first quadrant ; total 12 maximas on the circle}}$



For a point P on the circle shown in figure, the path difference in the two waves at P is $\Delta x = S_1P - S_2P = d\cos\theta = 3\lambda\cos\theta$

We know for constructive interference at P, the path difference must be an integral multiple of wavelength λ. Thus for a maxima at P, we have

$$3\lambda\cos\theta = 0;\ 3\lambda\cos\theta = \lambda;\ 3\lambda\cos\theta = 2\lambda;\ 3\lambda\cos\theta = 3\lambda$$

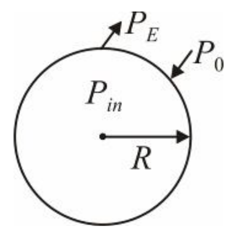
or $\quad \theta = \frac{\pi}{2}$ or $\theta = \cos^{-1}\left(\frac{1}{3}\right)$ or $\theta = \cos^{-1}\left(\frac{2}{3}\right)$ or $\theta = 0$

There are total 12 points of maxima on the circle

93. **[NA]**

Before expansion, $\quad P_{in} = P_0 + \frac{4T}{r}$

If charge q is given to the soap bubble, its potential $V = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{R}$



After expansion of bubble at radius R, the inside air pressure will be

$$P_{in} = P_0 - \frac{\sigma^2}{2\epsilon_0} + \frac{4T}{R}$$

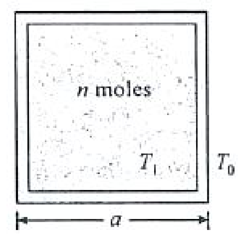
$$\left[\text{here } \sigma = \frac{q}{4\pi R^2} = \frac{\epsilon_0 VR}{R^2} = \frac{\epsilon_0 V}{R}\right]$$

By gas law $\quad P_1V_1 = P_2V_2$

$$\left(P_0 + \frac{4T}{r}\right)\left(\frac{4}{3}\pi r^3\right) = \left(P_0 - \frac{\sigma^2}{2\epsilon_0} + \frac{4T}{R}\right)\left(\frac{4}{3}\pi R^3\right)$$

$$\Rightarrow \quad P_0(R^3 - r^3) + 4T(R^2 - r^2) - \frac{\epsilon_0 V^2 R}{2} = 0$$

94. $\boxed{\boldsymbol{T_0 + (T_1 - T_0)e^{-\frac{6ka^2(\gamma-1)}{nRx}t}}}$



As shown in figure if at time $t = 0$, gas temperature is T_1 and after time t, its temperature is T, heat will be conducted through the walls of container, the rate of heat conducted to outside surrounding can be written as

$$\frac{dQ}{dt} = \frac{k(6a^2)}{x}[T - T_0] \qquad \text{... (i)}$$

If dQ heat is conducted outside in a time dt and due to this if the temperature of gas falls by dT, we have

$$dQ = -nC_V dt \qquad \text{[As volume of container is constant]}$$

or $\quad dQ = -\frac{nR}{\gamma - 1}dT \quad \left[\text{as } C_V = \frac{R}{\gamma-1}\right] \qquad \text{... (ii)}$

From equation (i) and (ii) we have

$$\frac{nR}{\gamma-1}\frac{dT}{dt} = -\frac{6ka^2}{x}(T - T_0) \text{ or } \frac{dT}{T - T_0} = -\frac{6ka^2(\gamma-1)}{nRx}dt$$

Integration the above expression within proper limits from beginning $(t = 0)$, we get

$$T = T_0 + (T_1 - T_0)e^{-\frac{6ka^2(\gamma-1)}{nRx}t}$$

95. $\boxed{\pi\left[\sqrt{\frac{\rho l}{\rho_1 g}}+\sqrt{\frac{\rho l}{\rho_2 g}}\right]}$

In equilibrium position, net force on pole is zero.

$$\rho Alg = \rho_1 Al_1 g \qquad \ldots\text{(i)}$$

In displaced position, net force, $F_1 = \rho Alg - \rho_1 Al_1 g - \rho_2 Ayg = -\rho_2 Agy$

Acceleration of the pole, $a_1 = -\frac{\rho_2 Ag}{\rho Al} y$

Hence in the liquid of density $\rho_2, \omega_1 = \sqrt{\frac{\rho_2 g}{\rho l}} \Rightarrow T_1 = 2\pi\sqrt{\frac{\rho l}{\rho_2 g}}$

The pole system is in the liquid of density ρ_2 for half time period only. When it moves in the liquid of density ρ_1, the net force is $F_2 = -\rho Alg + (l_1 - y')A\rho_1 g = -\rho_1 Agy'$

$$a_2 = -\frac{\rho Ag}{\rho Al} y' \Rightarrow \omega_2 = \sqrt{\frac{\rho_1 g}{\rho l}}$$

$$T_2 = 2\pi\sqrt{\frac{\rho l}{\rho_1 g}}$$

Total time period of oscillation of the cylinder is $T = \frac{T_1}{2} + \frac{T_2}{2} = \pi\left[\sqrt{\frac{\rho l}{\rho_1 g}} + \sqrt{\frac{\rho l}{\rho_2 g}}\right]$

96. $\boxed{\frac{V_0}{d}}$

Path difference $= (\mu - 1)d$

$$\mu = \frac{V_0}{V'} = \frac{\sqrt{\frac{\gamma R(T_0)}{m_0}}}{\sqrt{\frac{\gamma R(4T_0)}{m_0}}} = \frac{1}{2}$$

| Path difference | = $\left|\left(\frac{1}{2} - 1\right)d\right| = \frac{d}{2}$

For minima, path difference $= (2n+1)\frac{\lambda}{2} = \frac{d}{2}$

$$\frac{d}{(2n+1)} = \lambda = \frac{V_0}{f}$$

$$f = \frac{V_0(2n+1)}{d}$$

$$f_{min} = \frac{V_0}{d}$$

97. $\frac{p_a a^3}{RT_a}\left[1 + \left(\frac{T_a}{T_0} - 1\right)e^{-\frac{2T_a}{7rp_a a^3}t}\right]$

As the gas can leak out of the hole, the pressure inside the vessel will be equal to the atmospheric pressure p_a. Let n be the amount of the gas (moles) in the vessel at time *t*. Suppose an amount dQ of heat is given to the gas in time *dt*. Its temperature increases by *dT* where $dQ = nC_p dT$

If the temperature of the gas is T at time *t*, we have

$$\frac{dQ}{dt}=\frac{T_a-T}{r}$$

or, $\left(C_p r\right) ndT=\left(T_a-T\right)dt$ ....(i)

We know, $p_a a^3-nRT$

or, $n\ dT+T\ dn=0$

or, $n\ dT=-T\ dn$ ....(ii)

Also, $T=\dfrac{p_a a^3}{nR}$ ....(iii)

Using (ii) and (iii) in (i),

$$\frac{-C_p r\ p_a a^3}{nR}dn=\left(T_a-\frac{p_a\ a^3}{nR}\right)dt$$

or $$\frac{dn}{nR\left(T_a-\dfrac{p_a\ a^3}{nR}\right)}=-\frac{dt}{C_p\ r\ p_a\ a^3}$$

or $$\int_{n_0}^{n}\frac{dn}{nRT_a-p_a a^3}=-\int_0^t\frac{dt}{C_p\ r\ p_a a^3}$$

where $n_0=\dfrac{p_a a^3}{RT_0}$ is the initial amount of the gas in the vessel. Thus,

$$\frac{1}{RT_a}ln\frac{nRT_a-p_a a^3}{n_0 RT_a-p_a a^3}=-\frac{t}{C_p r\ p_a a^3}$$

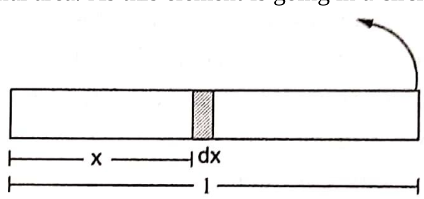
or, $$nRT_a-p_a a^3=\left(n_0 RT_a-p_a a^3\right)e^{-\frac{RT_a}{C_p r\ p_a a^3}t}$$

Writing $n_0=\dfrac{p_a a^3}{RT_0}$ and $C_p=C_v+R=\dfrac{7R}{2}$

$$n=\frac{p_a a^3}{RT_a}\left[1+\left(\frac{T_a}{T_0}-1\right)e^{-\frac{2T_a}{7rp_a a^3}t}\right]$$

98. **[NA]**

Consider an element of the gas between the cross sections at distances x and $x+dx$ from the fixed end (figure). If p be the pressure at x and $p+dp$ at $x+dx$, the force acting on the element towards the centre is Adp, where A is the cross sectional area. As this element is going in a circle of radius x.



$$Adp=\left(dm\right)\omega^2 x \quad ....(i)$$

where dm = mass of the element. Using $pV=nRT$ on this element,

$$pAdx=\frac{dm}{M}RT$$

or, $$dm=\frac{MpA}{RT}dx$$

Putting in (i),

$$Adp = \frac{MpA}{RT}\omega^2 x\,dx$$

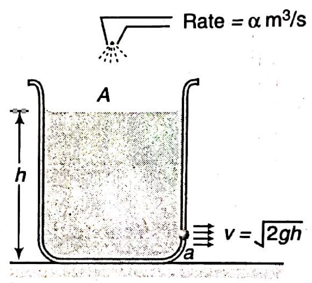
or, $$\int_{p_1}^{p_2}\frac{dp}{p} = \int_0^l \frac{M\omega^2}{RT}x\,dx$$

or, $$ln\frac{p_2}{p_1} = \frac{M\omega^2 l^2}{2RT}$$

or, $$p_2 = p_1 e^{\frac{M\omega^2 l^2}{2RT}}$$

99. **(i)** $\boxed{\dfrac{\alpha^2}{2ga^2}}$ **(ii)** $\dfrac{A}{ag}\left[\dfrac{\alpha}{a}\ln\left\{\dfrac{\alpha - a\sqrt{2gh}}{\alpha}\right\} - \sqrt{2gh}\right]$

(i) Level will be maximum when



Rate of inflow of water = rate of outflow of water

i.e., $\alpha = av$

or $\alpha = a\sqrt{2gh_{max}} \Rightarrow h_{max} = \dfrac{\alpha^2}{2ga^2}$

(ii) Let at time t, the level of water be h. Then,

$$A\left(\frac{dh}{dt}\right) = \alpha - a\sqrt{2gh} \text{ or } \int_0^h \frac{dh}{\alpha - a\sqrt{2gh}} = \int_0^t \frac{dt}{A}$$

Solving this, we get :

$$t = \frac{A}{ag}\left[\frac{\alpha}{a}ln\left\{\frac{\alpha - a\sqrt{2gh}}{\alpha}\right\} - \sqrt{2gh}\right]$$

100. **(i)** **[4R]** **(ii)** $\boxed{\sqrt{\dfrac{8\sigma}{\rho h r^2}}}$

(i) According to the first law of thermodynamics, the amount of heat transmitted dQ to the gas in the bubble is found as $dQ = nC_V dT + pdV$

where the molar heat capacity at arbitrary process is as follows $C = \dfrac{1}{n}\dfrac{dQ}{dt} = C_V + \dfrac{p}{n}\dfrac{dV}{dT}$

Here C_V stands for the molar heat capacity of the gas at constant volume, p designates its pressure, n is the total amount of moles of gas in the bubble, V and T denote the volume and temperature of the gas, respectively.

According to the Laplace formula, the gas pressure inside the bubble is defined by $p = \dfrac{4\sigma}{r}$

Thus, the equation of any equilibrium process with the gas in the bubble is a polytrope of the form

$p^3V = const$

The equation of state of an ideal gas has the form $pV = nRT$ and hence equation $T^3V^{-2} = const$

Differentiating the derivative with respect to temperature sought is found as $\dfrac{dV}{dT} = \dfrac{3V}{2T}$

Taking into account that the molar heat capacity of a diatomic gas at constant volume is $C_V = \dfrac{5}{2}R$

and using it is finally obtained that $C = C_V + \dfrac{3}{2}R = 4R$

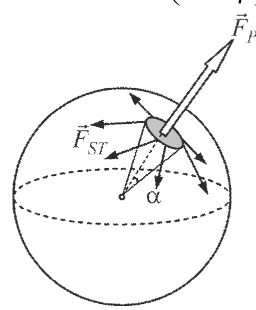
(ii) Since the heat capacity of the gas is much smaller than the heat capacity of the soap film, and there is heat exchange between them, the gas can be considered as isothermal since the soap film plays the role of thermostat. Consider the fragment of soap film, limited by the angle α as shown in the figure. It's area is found as : $S = \pi(ar)^2$

And the corresponding mass is obtained as $m = \rho Sh$.

Let x be an increase in the radius of the bubble, then the Newton second law for the fragment of the soap film mentioned above takes the form $m\ddot{x} = p'S' - F_{surf}$.

Where F_{surf} denotes the projection of the resultant surface tension force acting in the radial direction, p' stands for the gas pressure beneath the surface of the soap film and $S' = S\left(1 + 2\dfrac{x}{r}\right)$.

F_{surf} is easily found as $F_{surft} = F_{ST}\ \alpha = \sigma \cdot 2 \cdot 2\pi[r + x)\alpha] \cdot \alpha$



Since the gaseous process can be considered isothermal, it is written that $p'V' = pV$

Assuming that the volume increase is quite small,

$$p' = p\frac{1}{\left(1+\dfrac{x}{r}\right)^3} \approx p\frac{1}{\left(1+\dfrac{3x}{r}\right)} \approx P\left(1 - \frac{3x}{r}\right)$$

Thus, from (B . 10) – (B . 16) and (B . 3) the equation of small oscillations of the soap film is derived as $\rho h\ddot{x} = -\dfrac{8\sigma}{r^2}x$. With the frequency $\omega = \sqrt{\dfrac{8\sigma}{\rho h r^2}}$